

Classifying bent functions by their Cayley graphs

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Motivation

Question:

Which strongly regular graphs arise as Cayley graphs of bent Boolean functions?

Bent functions

Definition 1

The Walsh Hadamard transform of a Boolean function $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ is

$$W_f(x) := \sum_{y \in \mathbb{F}_2^{2m}} (-1)^{f(y) + \langle x, y \rangle}$$

Definition 2

*A Boolean function $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ is **bent** if and only if its Walsh Hadamard transform has constant absolute value 2^m .*

(Dillon 1974; Rothaus 1976)

The Cayley graph of a Boolean function

Definition 3

The *Cayley graph* $\text{Cay}(f)$ of a Boolean function

$$f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2 \quad \text{where} \quad f(0) = 0$$

is an undirected graph with

$$V(\text{Cay}(f)) := \mathbb{F}_2^n, \quad (x, y) \in E(\text{Cay}(f)) \Leftrightarrow f(x + y) = 1.$$

Strongly regular graphs

Definition 4

A simple graph Γ of order v is *strongly regular* with parameters (v, k, λ, μ) if

- ▶ each vertex has degree k ,
- ▶ each adjacent pair of vertices has λ common neighbours, and
- ▶ each nonadjacent pair of vertices has μ common neighbours.

(Brouwer, Cohen and Neumaier 1989)

Cayley graphs of bent functions

Proposition 1

(Bernasconi and Codenotti 1999)

The Cayley graph $\text{Cay}(f)$ of a bent function f on $\mathbb{F}_2^{2m}$ with $f(0) = 0$ is a strongly regular graph with $\lambda = \mu$.

The parameters of $\text{Cay}(f)$ are

$$(v, k, \lambda) = (4^m, 2^{2m-1} - 2^{m-1}, 2^{2m-2} - 2^{m-1})$$
$$\text{or } (4^m, 2^{2m-1} + 2^{m-1}, 2^{2m-2} + 2^{m-1}).$$

(Bernasconi and Codenotti 1999)

Extended affine equivalence

Definition 5

For bent functions $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$,

f is *extended affine equivalent* to g if and only if

$$g(x) = f(Ax + b) + \langle c, x \rangle + \delta$$

for some $A \in GL(2m, 2)$, $b, c \in \mathbb{F}_2^{2m}$, $\delta \in \mathbb{F}_2$.

(Budaghyan, Carlet and Pott 2006)

General linear equivalence

Definition 6

For bent functions $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$, f is *general linear equivalent* to g if and only if

$$g(x) = f(Ax)$$

for some $A \in GL(2m, 2)$.

Extended translation equivalence

Definition 7

For bent functions $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$,
 f is *extended translation equivalent* to g if and only if

$$g(x) = f(x + b) + \langle c, x \rangle + \delta$$

for $b, c \in \mathbb{F}_2^{2m}$, $\delta \in \mathbb{F}_2$.

Cayley equivalence

Definition 8

For $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$, with both f and g bent,
we call f and g **Cayley equivalent**, and write $f \equiv g$,
if and only if $f(0) = g(0) = 0$ and $\text{Cay}(f) \equiv \text{Cay}(g)$ as graphs.

Equivalently, $f \equiv g$ if and only if $f(0) = g(0) = 0$ and
there exists a bijection $\pi : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2^{2m}$ such that

$$g(x + y) = f(\pi(x) + \pi(y)) \quad \text{for all } x, y \in \mathbb{F}_2^{2m}.$$

Extended Cayley equivalence

Definition 9

For $f, g : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$, with both f and g bent,
if there exist $\delta, \epsilon \in \{0, 1\}$ such that $f + \delta \equiv g + \epsilon$,
we call f and g **extended Cayley (EC) equivalent** and write $f \cong g$.

Extended Cayley equivalence is an equivalence relation on the set of all bent functions on $\mathbb{F}_2^{2m}$.

General linear equivalence implies Cayley equivalence

Theorem 1

If f is bent with $f(0) = 0$ and $g(x) := f(Ax)$ where $A \in GL(2m, 2)$, then g is bent with $g(0) = 0$ and $f \equiv g$.

Proof.

$$g(x + y) = f(A(x + y)) = f(Ax + Ay) \quad \text{for all } x, y \in \mathbb{F}_2^{2m}.$$



Extended affine, extended translation, and extended Cayley equivalence (1)

Theorem 2

For $A \in GL(2m, 2)$, $b, c \in \mathbb{F}_2^{2m}$, $\delta \in \mathbb{F}_2$, $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$,
the function

$$h(x) := f(Ax + b) + \langle c, x \rangle + \delta$$

can be expressed as $h(x) = g(Ax)$ where

$$g(x) := f(x + b) + \langle (A^{-1})^T c, x \rangle + \delta,$$

and therefore if f is bent then $h \cong g$.

Extended affine, extended translation, and extended Cayley equivalence (2)

Therefore, to determine the extended Cayley equivalence classes within the extended affine equivalence class of a bent function $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$, for which $f(0) = 0$, we need only examine the extended translation equivalent functions of the form

$$f(x + b) + \langle c, x \rangle + f(b),$$

for each $b, c \in \mathbb{F}_2^{2m}$.

Weights and weight classes

Definition 10

The *weight* of a binary function is the cardinality of its *support*.
For f on $\mathbb{F}_2^{2m}$

$$\text{supp}(f) := \{x \in \mathbb{F}_2^{2m} \mid f(x) = 1\}.$$

A bent function f on $\mathbb{F}_2^{2m}$ has weight

$$\text{wt}(f) = 2^{2m-1} - 2^{m-1} \quad (\text{weight class } \text{wc}(f) = 0), \text{ or}$$

$$\text{wt}(f) = 2^{2m-1} + 2^{m-1} \quad (\text{weight class } \text{wc}(f) = 1).$$

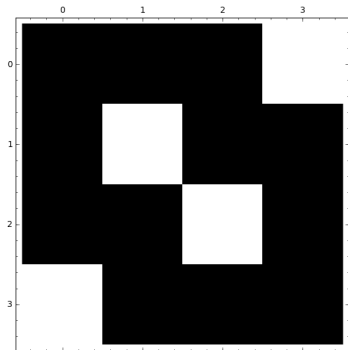
Quadratic bent functions have two extended Cayley classes

Theorem 3

For each $m > 0$, the extended affine equivalence class of quadratic bent functions $q : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2$ contains exactly two extended Cayley equivalence classes, corresponding to the two possible weight classes of $x \mapsto q(x + b) + \langle c, x \rangle + q(b)$.

For 2 dimensions: ET class $[f_{2,1}]$

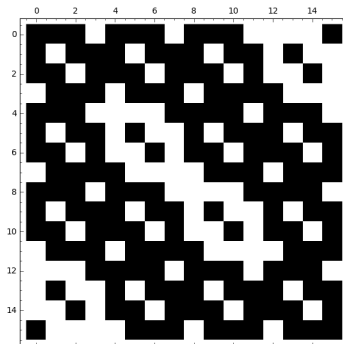
One extended affine class, containing the extended translation class $[f_{2,1}]$, where $f_{2,1}(x) := x_0x_1$.



$[f_{2,1}]$: 2 extended Cayley classes

For 4 dimensions: ET class $[f_{4,1}]$

One extended affine class, containing the extended translation class $[f_{4,1}]$, where $f_{4,1}(x) := x_0x_1 + x_2x_3$.



$[f_{4,1}]$: 2 extended Cayley classes

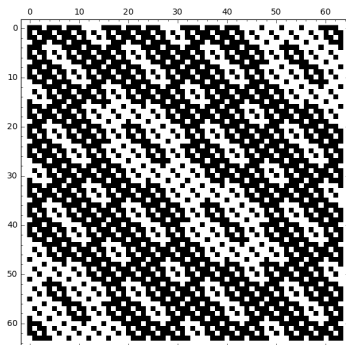
For 6 dimensions: ET classes

Four extended affine classes, containing the following extended translation classes:

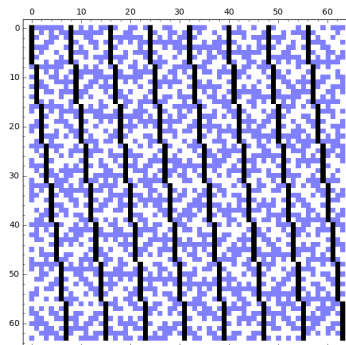
Class	Representative
$[f_{6,1}]$	$f_{6,1} := x_0x_1 + x_2x_3 + x_4x_5$
$[f_{6,2}]$	$f_{6,2} := x_0x_1x_2 + x_0x_3 + x_1x_4 + x_2x_5$
$[f_{6,3}]$	$f_{6,3} := x_0x_1x_2 + x_0x_1 + x_0x_3 + x_1x_3x_4 + x_1x_5$ $+ x_2x_4 + x_3x_4$
$[f_{6,4}]$	$f_{6,4} := x_0x_1x_2 + x_0x_3 + x_1x_3x_4 + x_1x_5 + x_2x_3x_5$ $+ x_2x_3 + x_2x_4 + x_2x_5 + x_3x_4 + x_3x_5$

(Rothaus 1976; Tokareva 2015)

ET classes $[f_{6,1}]$ and $[f_{6,2}]$

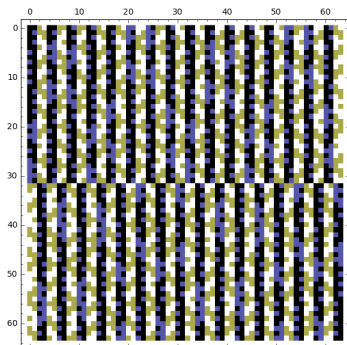


$[f_{6,1}]$: 2 extended Cayley classes

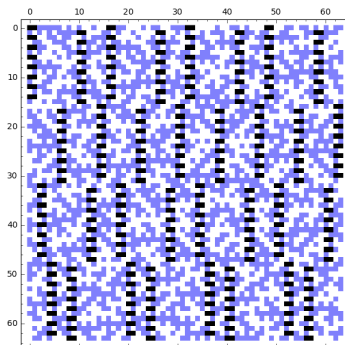


$[f_{6,2}]$: 3 extended Cayley classes

ET classes $[f_{6,3}]$ and $[f_{6,4}]$



$[f_{6,3}]$: 4 extended Cayley classes



$[f_{6,4}]$: 3 extended Cayley classes

For 8 dimensions: number of bent functions and EA classes

According to Langevin and Leander (2011) there are
 $99\,270\,589\,265\,934\,370\,305\,785\,861\,242\,880 \approx 2^{106}$ bent functions in
dimension 8.

The number of EA classes has not yet been published, let alone a
list of representatives.

(Langevin and Leander 2011)

For 8 dimensions, up to degree 3: extended translation classes

Ten extended affine classes are listed in Braeken's thesis (2006), containing the following extended translation classes:

Class	Representative
$[f_{8,1}]$	$f_{8,1} := x_0x_1 + x_2x_3 + x_4x_5 + x_6x_7$
$[f_{8,2}]$	$f_{8,2} := x_0x_1x_2 + x_0x_3 + x_1x_4 + x_2x_5 + x_6x_7$
$[f_{8,3}]$	$f_{8,3} := x_0x_1x_2 + x_0x_6 + x_1x_3x_4 + x_1x_5 + x_2x_3 + x_4x_7$
$[f_{8,4}]$	$f_{8,4} := x_0x_1x_2 + x_0x_2 + x_0x_4 + x_1x_3x_4 + x_1x_5 + x_2x_3 + x_6x_7$
$[f_{8,5}]$	$f_{8,5} := x_0x_1x_2 + x_0x_6 + x_1x_3x_4 + x_1x_4 + x_1x_5 + x_2x_3x_5 + x_2x_4 + x_3x_7$
$[f_{8,6}]$	$f_{8,6} := x_0x_1x_2 + x_0x_2 + x_0x_3 + x_1x_3x_4 + x_1x_6 + x_2x_3x_5 + x_2x_4 + x_5x_7$
$[f_{8,7}]$	$f_{8,7} := x_0x_1x_2 + x_0x_1 + x_0x_2 + x_0x_3 + x_1x_3x_4 + x_1x_4 + x_1x_5 + x_2x_3x_5 + x_2x_4 + x_6x_7$
$[f_{8,8}]$	$f_{8,8} := x_0x_1x_2 + x_0x_5 + x_1x_3x_4 + x_1x_6 + x_2x_3x_5 + x_2x_4 + x_3x_7$
$[f_{8,9}]$	$f_{8,9} := x_0x_1x_6 + x_0x_3 + x_1x_4 + x_2x_3x_6 + x_2x_5 + x_3x_4 + x_4x_5x_6 + x_6x_7$
$[f_{8,10}]$	$f_{8,10} := x_0x_1x_2 + x_0x_3x_6 + x_0x_4 + x_0x_5 + x_1x_3x_4 + x_1x_6 + x_2x_3x_5 + x_2x_4 + x_3x_7$

(Braeken 2006; Tokareva 2015)

ET classes $[f_{8,5}]$ and $[f_{8,6}]$ are the same

Theorem 4

Functions $f_{8,5}$ and $f_{8,6}$ are linearly equivalent and therefore Braeken's ET classes $[f_{8,5}]$ and $[f_{8,6}]$ are the same class.

Proof.

Apply the permutation $\pi := (x_0 \ x_5 \ x_4)(x_1 \ x_2 \ x_3)(x_6 \ x_7)$ to

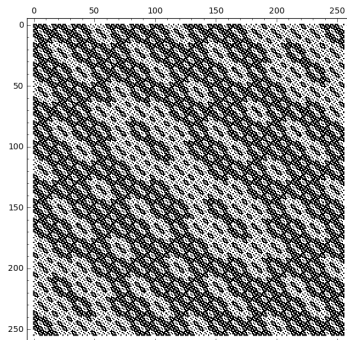
$$f_{8,5} = x_0x_1x_2 + x_0x_6 + x_1x_3x_4 + x_1x_4 + x_1x_5 + x_2x_3x_5 + x_2x_4 + x_3x_7$$

to obtain

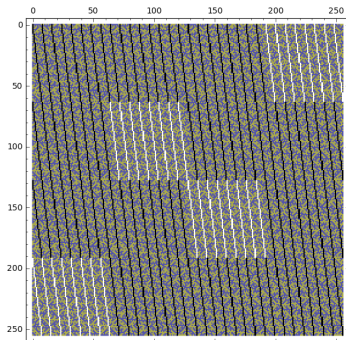
$$\begin{aligned} \pi(f_{8,5}) &= x_5x_2x_3 + x_5x_7 + x_2x_1x_0 + x_2x_0 + x_2x_4 + x_3x_1x_4 + x_3x_0 + x_1x_6 \\ &= x_0x_1x_2 + x_0x_2 + x_0x_3 + x_1x_3x_4 + x_1x_6 + x_2x_3x_5 + x_2x_4 + x_5x_7 \\ &= f_{8,6}. \end{aligned}$$



ET classes $[f_{8,1}]$ and $[f_{8,2}]$



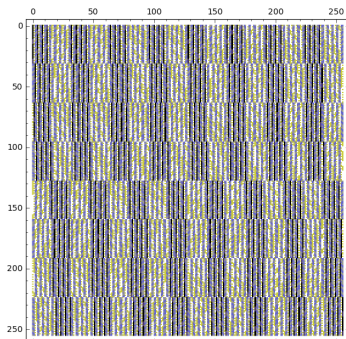
$[f_{8,1}]$: 2 extended Cayley classes



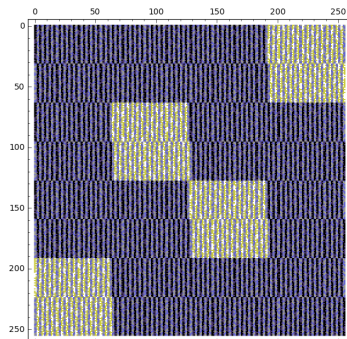
$[f_{8,2}]$: 4 extended Cayley classes

Graphs for class 0 of $[f_{8,1}]$ and class 0 of $[f_{8,2}]$ are isomorphic.

ET classes $[f_{8,3}]$ and $[f_{8,4}]$

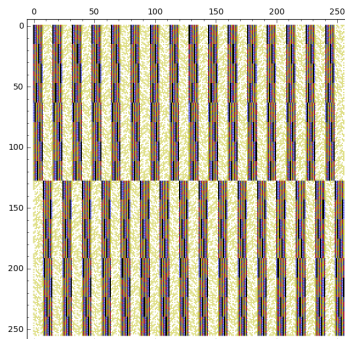


$[f_{8,3}]$: 6 extended Cayley classes



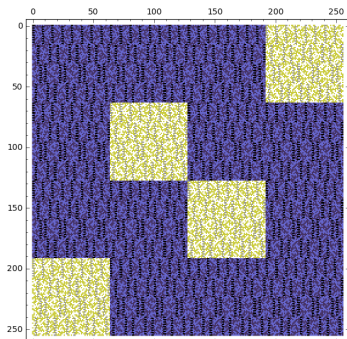
$[f_{8,4}]$: 5 extended Cayley classes

ET class $[f_{8,5}]$

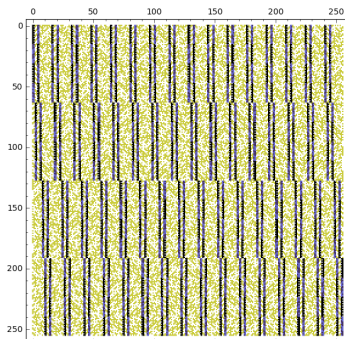


$[f_{8,5}]$: 9 extended Cayley classes

ET classes $[f_{8,7}]$ and $[f_{8,8}]$

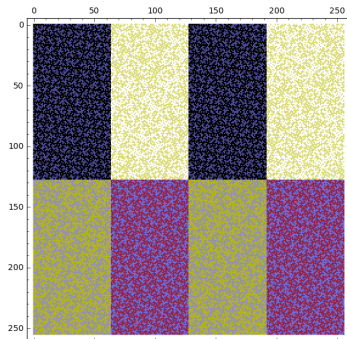


$[f_{8,7}]$: 5 extended Cayley classes

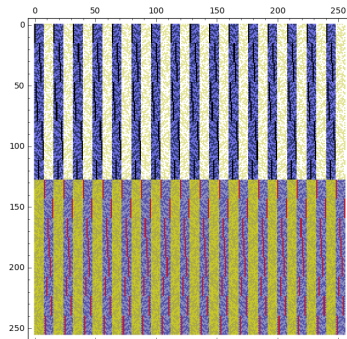


$[f_{8,8}]$: 6 extended Cayley classes

ET classes $[f_{8,9}]$ and $[f_{8,10}]$



$[f_{8,9}]$: 8 extended Cayley classes



$[f_{8,10}]$: 10 extended Cayley classes

Preprint, source code and documentation

Preprint: Paul Leopardi, Classifying bent functions by their Cayley graphs, arXiv:1705.04507 [math.CO]. Revised, December, 2018.

CoCalc: Public worksheets, Sage and Python source code

<http://tinyurl.com/Boolean-Cayley-graphs>

GitHub: Sage and Python source code

<https://github.com/penguian/Boolean-Cayley-graphs>

SourceForge: Documentation

<https://boolean-cayley-graphs.sourceforge.io/>

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SageMath, CoCalc, Bliss, Nauty, MPI4py, SQLite3, DB Browser for SQLite, PostgreSQL, Psycopg2.